

COMP 3311: Database Management Systems

Movie Streaming Management System

E-R Schema Design Assignment: Application Description

Reelflics Movie Streaming Service LLC (Reelflics) wants to develop an information system to manage a movie streaming service. Data about members, movies, cast members (i.e., actors and actresses) and directors of those movies should be maintained.

Reelflics employees should be able to create and modify movie, cast member and director information. Basic movie information should include a title, synopsis, one or more genres, release year, running time and MPAA rating. The MPAA rating should have one of the following values: G (all ages), PG (parental guidance), PG-13 (parents strongly cautioned), R (under 17 needs adult), or NC-17 (no one under 17 admitted). Additional information that should be maintained for a movie includes its IMDB rating (a decimal number between 1 and 10), if any, its Reelflics rating, if any, the main cast members that appear in the movie and the role (or roles) that each cast member plays in the movie, and the directors of the movie. The Reelflics rating for a movie should be the average of the individual ratings given by members.

Information maintained about cast members and directors should include name, biography, gender, birthdate, if known, and date of death, if any. Only information of cast members and directors that are associated with at least one movie that Reelflics streams should be maintained. While a movie could have several cast members, it also might have no cast members. However, every movie should have at least one director and could have several directors. Moreover, a director of a movie could also be a cast member of the movie.

Only the following academy award information should be maintained for a movie:

- whether the movie won the best picture award.
- whether any of the cast members won the best actor, best actress, best supporting actor or best supporting actress award.
- whether the directors won the best director award.

While cast members could have won multiple acting awards, they cannot win the same acting award more than once for a given movie. Moreover, only male cast members can win the best actor or best supporting actor award and only female cast members can win the best actress or best supporting actress award.

To be able to stream movies on Reelflics, a person should register as a member and should provide all of the following information: a unique username, a unique pseudonym (used for reviews as described subsequently), first (given) name, last (family) name, occupation, email, gender, birthdate, phone number and education level. Members should be charged a monthly membership fee. Therefore, each member should provide the following information for one credit card, to which the membership fee should be charged: card number, cardholder name, card type, security number/code, expiry month and expiry year. It should be possible to pay for multiple memberships with the same credit card. Only the following credit card types should be accepted: American Express, MasterCard, UnionPay and Visa. A member should be able to modify any of their information except for the username and pseudonym.

A member should be able to stream the same movie multiple times. Each time a movie is streamed, the date and time that the member started streaming the movie should be recorded. A member should be able to place movies that they want to watch on a watchlist. A member's movie watchlist can be empty or contain many movies. Once a movie is placed on a member's movie watchlist, it should remain there until it is removed by the member.

Members should be able to provide movie reviews consisting of a review title, an integer rating between 1 and 10, some review text and the review date. To protect members' privacy, reviews should show only members' pseudonym and not their real name. Members should be able to review only movies that they have streamed. A member should be able to provide at most one review for a movie and should be able to edit their reviews.